# AI TOOLS FOR RESEARCH

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## Phase II — Experiment Plan

**Research topic:** Comparing AI-Assisted and Manual Data Labelling Accuracy on Scientific Datasets

## Prompt 06: Verify Datasets and Comparators

### Research question and dataset

At the same verification budget, does uncertainty-based selection of scientific citation-intent predictions recover more errors than random selection? The selected dataset is **SciCite** (Cohan et al., 2019), containing scientific citation contexts with three labels: background, method and result [1]. The official archive is available from the original SciCite distribution. Its JSONL records use **string** for text, **label** for class, and **unique_id** and **citingPaperId** for identity checks.

### Dataset structure and verification

The recorded dataset check found 8,194 training records, 916 validation records and 1,859 test records after unique-ID handling, with no missing text. These counts came from the previously executed Colab audit. Use the official training, development and test partitions; learn the TF-IDF vocabulary from training data only. The dataset archive and exact data-use terms must be checked at its source before redistributing any raw records.

### Baselines and fair comparison

Baseline 1 is TF-IDF (unigrams and bigrams, at most 20,000 features) with logistic regression. Baseline 2 is the same TF-IDF representation with a linear support vector machine. These are computationally affordable, interpretable comparisons, **not** exact reproductions of the original SciCite neural method. The reference label serves as an oracle for simulated correction; this is not a trial with human reviewers.

### Protocol and risks

Use the same training examples, feature settings and validation split for both baselines; report accuracy and macro-F1. Audit repeated text, repeated IDs and paper-level overlap before evaluation. The earlier audit found two exact training–validation context overlaps and shared citing-paper IDs across splits. Remove the two exact overlapping contexts in a separately reported validation analysis (914 examples), while retaining the official-split results for comparability. A paper-disjoint evaluation subset is useful as a sensitivity check, but does not replace fully group-disjoint retraining.

**Independent checks:** verify the original dataset citation and acquisition location; inspect its field names and label definitions; confirm reuse terms; confirm notebook dependency versions and resource requirements; check that no test labels enter model fitting or model selection.

## Prompt 07: Experiment Blueprint

### Input, output and modules

**Input:** a SciCite citation context and its reference class. **Model output:** predicted citation class and a vector of SVM decision scores. **Verification output:** a ranked queue of examples for correction and the resulting evaluated labels.

**Processing flow:** Official SciCite JSONL → validate schema and splits → fit TF-IDF on train → train logistic regression and LinearSVC → evaluate on validation → rank SVM predictions by lowest top-two decision-score margin → simulate correction from reference labels → compare against equal-budget random selection → evaluate fixed choice on held-out test → paper-overlap sensitivity check.

### Metrics and controls

**Primary:** corrected-label accuracy (higher is better). **Secondary:** macro-F1 and errors recovered (higher is better), plus baseline class-wise precision/recall. Keep the same fitted SVM and the same examples for random

and uncertainty selection, with an equal number reviewed. Repeat random selection 100 times using NumPy seed 42 and report mean and standard deviation. An SVM margin is a ranking score, not a calibrated probability.

**Budget and evaluation design**

The recorded validation exploration used review budgets 0, 10, 25, 50, 100, 150, 200 and 300. The subsequent held-out comparison used 100 reviews among 1,859 test examples. A sensitivity check excluded test examples with citing-paper IDs present in training and used 87 reviews among 1,624 examples to preserve approximately the same review proportion. These details are documented **retrospectively** from the actual executed work; the study was not preregistered before inspecting validation results.

**Deeper analysis, reproducibility and limitations**

Generate a confusion matrix, save misclassified examples, plot accuracy and errors recovered against review budget, and perform the citing-paper-overlap sensitivity analysis. Save per-run CSV files, figures, the notebook, package versions, seeds and commands. Do not select or revise the model using held-out test scores. If later protocol changes are needed, record the reason, date and impact in research_decisions.md.

Limitations include potentially incorrect reference labels, imperfect simulation of human corrections, shared citing papers, a single domain and dependence on one uncertainty score. An oracle experiment can estimate selection efficiency but cannot establish human correction speed, labour cost or automation bias.

## References

[1] Cohan, A., Ammar, W., van Zuylen, M., and Cady, F. (2019). Structural Scaffolds for Citation Intent Classification in Scientific Publications. NAACL-HLT.

[2] Settles, B. (2009). Active Learning Literature Survey. University of Wisconsin–Madison, Technical Report 1648.

[3] Nikam, U. Comparing AI-Assisted and Manual Data Labelling Accuracy on Scientific Datasets (previous literature review), Sections 3–6.